

Windows 11 System Software Investigation

LabHands-On Exploration Activity

Lab Overview

Objective: Students will explore Windows 11 system software to understand operating systems, device drivers, and system utilities through hands-on investigation.

Duration: 35 minutes **Prerequisites:** Windows 11 computers with standard user accounts **Safety Note:** All activities are read-only exploration - no system changes will be made

Part 1: Operating System Deep Dive

Activity 1A: System Information Discovery

Step 1: Access System Information

Method 1: Settings App

1. Press **Windows key + I** to open Settings
2. Click "**System**" in the left sidebar
3. Scroll down and click "**About**" at the bottom

Method 2: Classic System Information

1. Press **Windows key + R** to open Run dialog
2. Type **msinfo32** and press Enter
3. Wait for System Information window to open

Step 2: Complete the System Discovery Worksheet (6 minutes)

SYSTEM INFORMATION DISCOVERY WORKSHEET

None

BASIC SYSTEM INFORMATION:

Device Name: -----

Windows Edition: -----

Windows Version: -----

Build Number: -----

HARDWARE SPECIFICATIONS:

Processor (CPU): -----

Installed RAM: -----

System Type: -----

Storage Available: -----

SYSTEM DETAILS (from msinfo32):

BIOS Version: -----

Hardware Abstraction Layer: -----

Total Physical Memory: -----

Available Physical Memory: -----

DISCOVERY QUESTIONS:

1. What does "Build Number" tell you about your Windows version?

2. How much of your total RAM is currently available?

3. What company made your computer's processor?

Discussion Questions:

1. Why might the 'Available Physical Memory' be less than 'Total Physical Memory'?

2. What do you think 'Hardware Abstraction Layer' means?

Activity 1B: Built-in vs. User-Installed Software Hunt

Step 1: Explore Built-in Windows Apps (4 minutes)

Access Windows Apps:

1. Click **Start Menu**
2. Click **"All Apps"**
3. Look for apps that came with Windows 11

Find and Document Built-in Apps:

None

BUILT-IN WINDOWS 11 APPS CHECKLIST:

- | | | |
|---|---|-----------------------------------|
| ☐ Calculator | ☐ Camera | ☐ Clock |
| ☐ Mail | ☐ Maps | ☐ Weather |
| ☐ Microsoft Store | ☐ Paint | ☐ Notepad |
| ☐ Snipping Tool | ☐ Voice Recorder | ☐ Photos |
| ☐ Windows Security | ☐ File Explorer | ☐ Settings |

Additional built-in apps you discovered:

Part 2: Device Driver Detective Work

Activity 2A: Device Manager Exploration

Step 1: Access Device Manager

Method 1: Right-click Start Menu

1. Right-click the **Start** button
2. Select **"Device Manager"** from the menu

Method 2: Search Method

1. Press **Windows** key
2. Type **"Device Manager"**
3. Click on Device Manager when it appears

Step 2: Driver Investigation Worksheet

DEVICE DRIVER DETECTIVE WORKSHEET

None

DRIVER CATEGORY EXPLORATION:
(Click the arrow to expand each category and explore)

DISPLAY ADAPTERS:

Device Name: -----

Driver Version: -----

Driver Date: -----

What does this driver control? -----

AUDIO INPUTS AND OUTPUTS:

Device Name: -----

Driver Version: -----

Driver Date: -----

What does this driver control? -----

NETWORK ADAPTERS:

Device Name: -----

Driver Version: -----

Driver Date: -----

What does this driver control? -----

HUMAN INTERFACE DEVICES:

List 3 devices you found:

1. -----

2. -----

3. -----

CRITICAL THINKING:

1. What do you think would happen if the display adapter driver was missing?

2. Why do you think there are so many "Human Interface Devices"?

Safety Reminder: "We are only LOOKING at drivers, not changing anything. Do not click any buttons that say 'Update' or 'Uninstall'!"

Activity 2B: Driver Properties Deep Dive

Step 1: Examine Driver Details

Instructions:

1. In Device Manager, expand **"Display adapters"**
2. **Right-click** on your graphics card
3. Select **"Properties"**
4. Click the **"Driver"** tab

Step 2: Driver Analysis

DRIVER PROPERTIES INVESTIGATION:

None

DETAILED DRIVER INFORMATION:

Driver Provider: -----

Driver Date: -----

Driver Version: -----

Digital Signer: -----

DRIVER FILES:
(Click "Driver Details" button)

How many driver files are listed? -----

What file extension do most driver files have? ---

ROLLBACK FEATURE:
Is "Roll Back Driver" button available? -----

What do you think this feature does?

DRIVER UPDATES:

What happens when you click "Update Driver"?
(DON'T ACTUALLY CLICK - just predict)

Part 3: System Utilities Testing Lab

Activity 3A: Task Manager Process Analysis

Step 1: Access Task Manager

Method 1: Keyboard Shortcut

- Press **Ctrl + Shift + Esc**

Method 2: Right-click Taskbar

- Right-click empty space on taskbar
- Select **"Task Manager"**

Step 2: Process Investigation (3 minutes)

TASK MANAGER EXPLORATION WORKSHEET

None

PROCESSES TAB ANALYSIS:

Current running processes: -----

(Look at the number in parentheses next to "Processes")

TOP 5 CPU-USING PROCESSES:

1. -----

2. -----

3. -----

4. -----

5. -----

MEMORY USAGE ANALYSIS:

Total Memory Usage: -----

Available Memory: -----

PROCESS CATEGORIES:

- ☐ Apps (programs you opened)
- ☐ Background processes (system running automatically)
- ☐ Windows processes (operating system components)

DISCOVERY QUESTIONS:

1. Which category has the most processes running?

2. What processes are using the most memory?

3. Are there any processes you don't recognize?

Activity 3B: System Cleanup Utilities Exploration

Step 1: Access Disk Cleanup (1 minute)

1. Press **Windows key + R**
2. Type **cleanmgr** and press Enter
3. Select your main drive (usually C:)
4. Wait for scan to complete

Step 2: Cleanup Analysis

DISK CLEANUP INVESTIGATION:

None

CLEANUP CATEGORIES FOUND:

- ☐ Temporary files ☐ Recycle Bin
- ☐ Temporary Internet Files ☐ Thumbnails
- ☐ Downloaded Program Files ☐ System error memory dump files

SPACE ANALYSIS:

Total space can be freed: -----

Largest cleanup category: -----

CLEANUP SAFETY:

Which items do you think are safe to delete?

Which items might you want to keep?

CRITICAL THINKING:

1. Why does the computer create temporary files?

2. How often do you think you should run disk cleanup?

Safety Note: "We're only LOOKING at what can be cleaned up. Don't click 'OK' to actually delete files unless your teacher says it's okay!"

Activity 3C: Windows Security Interface Tour

Step 1: Access Windows Security (1 minute)

1. Press **Windows key + I** to open Settings
2. Click **"Privacy & security"**
3. Click **"Windows Security"**
4. Click **"Open Windows Security"**

Step 2: Security Features Exploration (2 minutes)

WINDOWS SECURITY DASHBOARD:

None

PROTECTION AREAS AVAILABLE:

- ☐ Virus & threat protection ☐ Account protection
- ☐ Firewall & network protection ☐ App & browser control
- ☐ Device security ☐ Device performance & health
- ☐ Family options

CURRENT STATUS CHECK:

Virus & threat protection: -----

Firewall status: -----

Last scan: -----

REAL-TIME PROTECTION:

Is real-time protection ON or OFF? -----

What does real-time protection do?

QUICK SCAN OPTION:

What happens when you click "Quick scan"?
(Predict - don't actually run it)

CRITICAL THINKING:

1. Why is this security software built into Windows?

2. What would happen if you turned off the firewall?

Lab Wrap-Up and Reflection (2 minutes)